

SingleRAN

Base Station OMCH Self-recovery Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-07-30).

Technical Changes

None

Editorial Changes

Revised the description of the recovery process of some OMCH-related parameters. For details, see [5.1 Principles](#).

Revised certain descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to NR, NB-IoT, LTE FDD, LTE TDD, UMTS, and GSM.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the Base Station OMCH Self-recovery feature, including its technical principles, related features, network impacts, and engineering guidelines. It consists of the following functions: automatic version rollback, automatic configuration data rollback, transmission link fault rectification, automatic OMCH establishment, automatic recovery of OMCH running data, and OM emergency channel-based self-healing. There is no feature ID for RATs except NR.

RAT	Feature ID	Feature Name	Chapter/Section
NR	FBFD-020101	Reliability	4 Automatic Configuration Data Rollback 5 Automatic OM Channel Establishment 6.1 Automatic Version Rollback 6.2 Transmission Link Fault Rectification 6.3 Automatic Recovery of OMCH Running Data 6.4 OM Emergency Channel-based Self-Healing

3 Overview

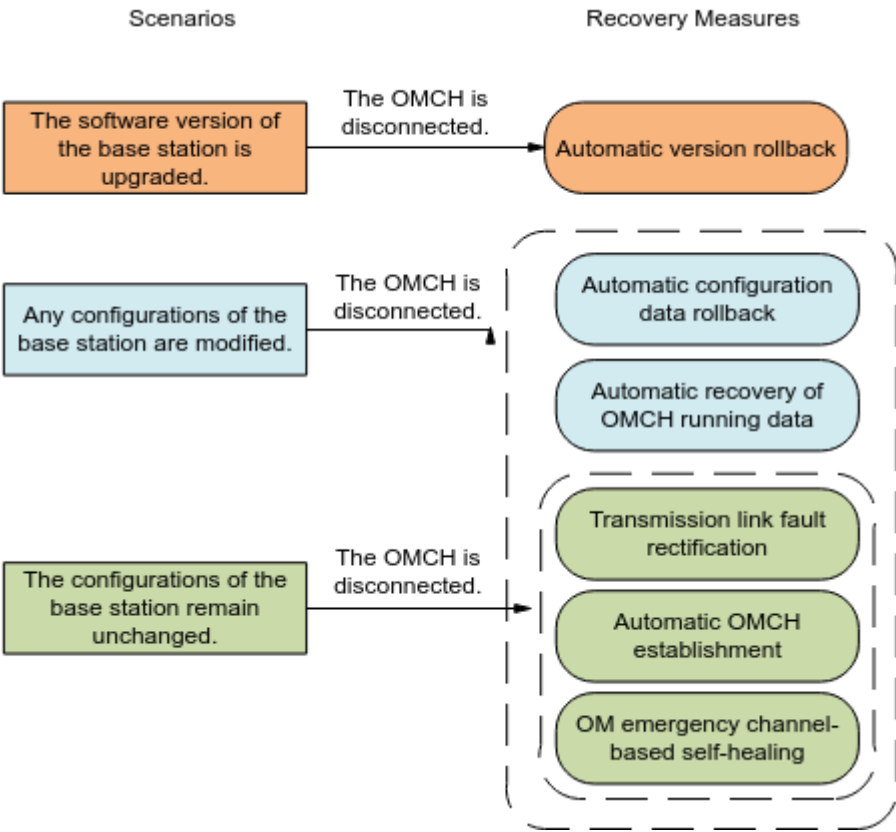
The traditional remote operation and maintenance channel (abbreviated as OM channel or OMCH) of a base station refers to the channel through which an eGBTS/NodeB/eNodeB/gNodeB communicates with the MAE or a GBTS communicates with a BSC to exchange operation and maintenance information about the base station.

If an OMCH is disconnected, measures are taken to recover the OMCH. These recovery measures include:

- Automatic version rollback
- Automatic configuration data rollback
- Transmission link fault rectification
- Automatic OMCH establishment
- Automatic recovery of OMCH running data
- OM emergency channel-based self-healing

The following figure shows the recovery measures in different scenarios.

Figure 3-1 Measures taken to recover the disconnected OMCH



Automatic version rollback: After an upgrade of base station software, the software version is automatically rolled back to the source version when rollback conditions are met, attempting to recover the OMCH.

Automatic configuration data rollback: The automatic configuration data rollback function is enabled before configuration data is modified. After the modification, if the OMCH continuously disconnects when the timer specifying the rollback time expires, the base station is automatically reset and rolls the configuration data back to the data before modification to recover the OMCH.

Transmission link fault rectification: When the OMCH and services are both continuously interrupted, the base station initiates a reset to recover the faulty OMCH.

Automatic OMCH establishment: When the function is enabled and automatic OMCH establishment conditions are met, the base station obtains the OMCH information through the DHCP Detect process and attempts to recover the OMCH.

Automatic recovery of OMCH running data: The OMCH running data collected while the OMCH works properly is used to restore the OMCH.

OM emergency channel-based self-healing: When the OM channel of a base station remains interrupted, users can manually enable the OM emergency channel-based self-healing function to attempt to recover the OM channel by resetting the base station and enabling the minimum configuration mode of the base station.

The Base Station OMCH Self-recovery feature minimizes the need of onsite handling in the case of an OMCH disconnection, and shortens the interruption time and saves costs.

4 Automatic Configuration Data Rollback

4.1 Principles

Automatic Configuration Data Rollback for eGBTS/NodeB/eNodeB/gNodeB

If you are concerned that inappropriate data modification may cause long OMCH disconnections and recovery failures (for example, when the IP address, IPsec address, or router is incorrectly modified), you are advised to enable the automatic configuration data rollback function before modifying the data related to the OMCH. This function can roll back the configuration data, which reduces the need for site visits.

NOTE

It is recommended that this function be disabled when the data related to the OMCH is locally modified. Otherwise, automatic configuration data rollback may be triggered as a result of an OMCH disconnection.

To enable this function, run the **CFM CB** command with **AUTORBKS** set to **ENABLE** and specify the **RBKTIME** and **RBKCONDITION** parameters.

When the **CFM CB** command is used to enable the automatic configuration data rollback function, the base station saves the current configuration data to the new configuration baseline (CB) created by this command and then the timer specified by **RBKTIME** starts. The data modification must be complete before the timer expires.

For the existing CB, you can also run the **MOD CB** command to enable the automatic configuration data rollback function. If the configuration cannot be complete when the timer specified by **RBKTIME** expires, run the **MOD CB** command to reconfigure the rollback parameters. If **AUTORBKS** is set to **ENABLE** and the **RBKTIME** parameter is reconfigured, the base station will restart the timing (the remaining rollback time can be queried by running the **DSP CB** command).

- The timer specified by **RBKTIME** has not expired.
If the OMCH is normal after its configuration data has been modified or recovers to the normal state after an interruption, and users do not require

rollback regardless of an OMCH interruption, run the **MOD CB** command to disable the automatic configuration data rollback function. This prevents the rollback of the configuration data in case of an OMCH disconnection.

 NOTE

Incorrect modification of certain data will not lead to an immediate OMCH disconnection because the modification must be first renegotiated with the peer NE. Such data includes parameters in **IPSECPROPOSAL**, **IKEPROPOSAL**, and **IKECFG** MOs. The settings of these parameters must be consistent with those of the peer NE.

If the base station resets before the timer specified by **RBKTIME** expires, the automatic configuration data rollback function still takes effect after the reset.

- The timer specified by **RBKTIME** expires.
 - If rollback conditions specified by **RBKCONDITION** are met, the base station resets, data configuration is rolled back to the corresponding CB used to enable this function, and ALM-26255 Automatic Version Rollback is reported. The base station will automatically disable this function after the rollback is complete.

 NOTE

An OMCH interruption can be detected within 5 minutes. If the OMCH is interrupted, the OMCH is considered faulty.

- If the rollback conditions are not met, the base station waits five more minutes before making a decision again. The remaining rollback time is 5 minutes. After the 5 minutes expire:
 - If rollback conditions specified by **RBKCONDITION** are met, the base station resets and data configuration is rolled back to the corresponding CB used to enable this function. If the rollback succeeds, the base station disables this function.
 - If rollback conditions are not met (for example, the OMCH is normal or has recovered to the normal state), the base station disables this function.
- If the **RBKCONDITION** parameter is set to **OMCH_AND_SERVICE_FAULT**, the automatic configuration data rollback function will be triggered only when the OMCH and all service channels are faulty.

 NOTE

If ALM-26255 Automatic Version Rollback is reported with the cause value being **Configuration Data**, this alarm was triggered by the automatic configuration data rollback function.

When the automatic configuration data rollback function is enabled on a batch of base stations, this function can be triggered for only the base stations with constant external transmission link interruptions once the timer specifying the rollback time expires.

For an OMCH disconnection caused by the batch configuration of base stations, if the MAE does not finish the configuration when the timer specified by **RBKTIME** expires, this function is triggered after the batch configuration is finished, if rollback conditions specified by **RBKCONDITION** are still met.

When users run the **ADD BRD** command to add a board during the batch configuration of base stations, the board may be automatically activated. If the timer specifying the rollback time elapses, but the board has not yet been activated, the base station will determine whether to trigger the automatic configuration data rollback function based on the rollback conditions (**RBKCONDITION**) after the board has been activated.

If the automatic configuration data rollback function is manually or automatically disabled, this function needs to be reconfigured if it is required.

The configuration data of peripheral NEs may be modified when the configuration data of base stations is being modified. After the automatic configuration data rollback function has been successfully triggered, the configuration data of base stations is rolled back. At the same time, the data of peripheral NEs may also need to be rolled back to ensure a normal OMCH. Therefore, the data modification of peripheral NEs during this period and the rollback methods should be obtained.

Automatic Configuration Data Rollback for GBTS

The GBTS does not support this function.

4.2 Network Analysis

4.2.1 Benefits

None

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

No requirements

RF Modules

This function does not depend on RF modules.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 When to Use

Modifying the radio configuration parameters does not lead to OM channel disconnection, so the automatic configuration data rollback function does not need to be enabled.

However, modifying the configuration data related to OM channel may cause continuous OM channel disconnection, which cannot be remotely recovered. It is recommended that the automatic configuration data rollback function be enabled in this scenario.

It is recommended that this function be disabled when the data related to the OM channel is locally modified. Otherwise, automatic configuration data rollback may be triggered as a result of an OM channel disconnection.

Each time before enabling this function, you are advised to collect the modified data of peripheral NEs and corresponding rollback methods.

4.4.2 Precautions

After the system successfully triggers the automatic data configuration rollback function, the base station will reset, during which services are temporarily interrupted.

In a multimode base station with multiple modes sharing a single board, for example, a main control board, resetting the base station or the main control board may affect services of other modes.

4.4.3 Data Configuration

4.4.3.1 Data Preparation

MO	Parameter Name	Parameter ID	Setting Notes
N/A	Automatic Rollback Switch	<i>AUTORBKS</i> <i>W</i>	When only the radio configuration parameters are modified and the OM channel is not affected, turn off this switch. When the configuration data of base stations is modified, which may cause continuous OM channel disconnections and recovery failures, it is recommended that this switch be turned on.

MO	Parameter Name	Parameter ID	Setting Notes
N/A	Rollback Time	<i>RBKTIME</i>	Users should estimate in advance the required time for modifying the configuration in order to set appropriate rollback time. If this parameter is set to 30 minutes according to estimation, the modification must be complete within 30 minutes. If the modification cannot be complete within 30 minutes, run the MOD CB command to reconfigure the rollback time in advance. If this parameter is set to a value too large, the OM channel may be disconnected for a long time (services may also be interrupted for a long time). If this parameter is set to a value too small, the automatic configuration data rollback function cannot take much effect in data protection. Assume that this parameter is set to 240 minutes. If users incorrectly configure a parameter 20 minutes after this function is enabled and the OM channel and services are interrupted, the system will wait 220 minutes before triggering this function. Such a long interruption affects user experience.

MO	Parameter Name	Parameter ID	Setting Notes
N/A	Rollback Condition	<i>RBKCONDITIO N</i>	Set this parameter to OMCH_FAULT in most cases. If this parameter is set to OMCH_AND_SERVICE_FAULT , the automatic configuration data rollback function will be triggered after the rollback time (<i>RBKTIME</i>) elapses only when the OM channel and all service channels are faulty. If an OM channel disconnection occurs whereas services are normal, the automatic configuration data rollback function will not be triggered.

4.4.3.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
CFM CB: MODE=UNFORCED, NAME="Rollback CB 1", COMMENT="CB for Rollback", AUTORBKSW =
ENABLE, RBKTIME=30, RBKCONDITION=OMCH_FAULT;
MOD CB: NAME="Rollback CB 1", COMMENT="CB for Rollback", AUTORBKSW = ENABLE, RBKTIME=30,
RBKCONDITION=OMCH_FAULT;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
MOD CB: NAME="Rollback CB 1", COMMENT="CB for Rollback", AUTORBKSW = DISABLE;
```

4.4.3.3 Using the MAE-Deployment

Not supported

4.4.4 Activation Verification

Run the MML command **DSP CB** to query the values of **Automatic Roll Back Switch** and **Roll Back Remaining Time**. If the value of **Automatic Roll Back Switch** is **ENABLE(Enable)**, the automatic configuration data rollback function has been enabled.

4.4.5 Network Monitoring

None

5 Automatic OM Channel Establishment

5.1 Principles

The automatic remote OMCH establishment function allows a base station to recover the remote OMCH through the DHCP Detect process. The automatic remote OMCH establishment function works only in certain scenarios. You are advised to run the **CFM CB** or **MOD CB** command to enable automatic configuration data rollback before remotely modifying the OMCH-related parameters. For details, see [4 Automatic Configuration Data Rollback](#).

The DHCP Detect process is classified into:

1. Recovery process of some OMCH-related parameters. This process is supported by the base station by default.

This recovery does not affect services.

- Scenario: non-IPsec networking or IPsec networking where the DHCP channel and OMCH are not protected by IPsec
- Prerequisite: The OMCH is disconnected and services are normal.
- Scope: The Ethernet port (physical-layer parameters) and the local IP address and route (network-layer parameters) of the OMCH can be restored.
 - If a route to the network segment defined by the peer IP address and peer subnet mask in the **DSP DHCPRLT** command output is not configured, the base station will automatically add such a route with the destination IP address and subnet mask being the peer IP address and peer subnet mask in this command.
 - If a network segment route defined by the peer IP address and peer subnet mask in the **DSP DHCPRLT** command output is configured but the OMCH cannot be restored using this route, the base station adds a route. The destination IP address of the route is the peer IP address in the **DHCHRLT** MO or the OSS IP address for the latest login, and the subnet mask is 255.255.255.255.
- Conditions: When services are normal but the OMCH is interrupted for 10 consecutive minutes (the wait time, namely the time interruption lasts, is

- 15 minutes if SSL authentication fails), automatic remote OMCH establishment will be triggered.
2. Recovery process of all OMCH-related parameters. This process is controlled by the **DHCPDW.SWITCH (LTE eNodeB, 5G gNodeB)** parameter.
- Scenario: IP transmission
 - Prerequisite: Both the OMCH and services are interrupted.
 - Scope: All parameters that are related to the OMCH at the physical layer, data link layer, and network layer can be recovered. This type of recovery requires detection on all ports.
 - Conditions: If any of the following conditions is met when the automatic remote OMCH establishment switch (specified by the **DHCPDW.SWITCH (LTE eNodeB, 5G gNodeB)** parameter) is turned on, automatic remote OMCH establishment will be triggered after the OMCH and service link have been interrupted for 10 consecutive minutes (the wait time, namely the time interruption lasts, is 15 minutes if SSL authentication fails):
 - The OMCH is disconnected after the **OMCH** MO is modified.
 - The OMCH is disconnected after configurations other than the **OMCH** MO are modified, and the transmission link fault rectification is triggered.
 - An unexpected reset unrelated to transmission links occurs, such as a power-on reset or upgrade-led reset.
 - The interruption lasts longer than the value specified by the **DHCPDETECT.DELAYTIME** parameter in the **SET DHCPDETECT** command.

 NOTE

- The **SET DHCPDETECT** command is mainly used in the transmission network reconstruction scenario. Before the reconstruction, this command must be executed to set **Detect Delay Time**, which is the recovery time reserved when the transmission reconstruction is interrupted.
 - During the reconstruction, the OMCH will be recovered after the Detect delay time elapses no matter whether the base station configuration is modified.
 - The **SET DHCPDETECT** command is a maintenance command and will not take effect after the base station is restarted. This command is not recommended for routine maintenance.
- The **SET DHCPDW** command has been used to set **DHCPDW.DELAYSW (LTE eNodeB, 5G gNodeB)** to **ENABLE(Enable)**, and the OMCH and services have been interrupted for two hours.

 NOTE

- The delayed reestablishment switch is used when the OMCH and services are both interrupted during routine Ethernet transmission maintenance.
- The automatic remote OMCH establishment triggered by the delayed reestablishment switch is only initiated on the Ethernet port after both the OMCH and services have been interrupted for two hours. The two-hour duration is reserved for recovery in case of network interruptions and is not configurable.

Restrictions in IPsec networking:

- When this function takes effect, IPsec Redundancy Among Multiple SeGWs only allows the base station to set up a IPsec tunnel to the primary SeGW, and IPsec tunnel backup is not supported. For details, see *IPsec*.
- This function supports IPsec networking for transmission ports with **INTERFACE.PT** set to **ETH**.
- This function supports only the IKE and IPsec negotiation algorithm suites used for establishing a temporary IPsec tunnel during automatic OMCH establishment. For details, see *Automatic OMCH Establishment*.
- In scenarios where certificates are used for IPsec authentication, the certificates can only be downloaded through the interface IP used by IKE for transmission.

Risks and impact of the OMCH recovery process:

- After the base station re-establishes the remote OMCH based on the configuration data obtained from the DHCP server, the running data is inconsistent with the configuration data. In this case, the base station reports ALM-25902 Remote Maintenance Link Running Data and Configuration Mismatch. Services can be restored only after the alarm is cleared. For details about how to clear the alarm, see *3900 & 5900 Series Base Station Alarm Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- The base station checks whether service-related transmission configurations need to be restored every 45 minutes.
 - If no transmission-related O&M operations are performed within 45 minutes, the base station may attempt to restore service-related transmission configurations. During this process, the OMCH will be temporarily interrupted.
 - If transmission-related O&M operations are performed within 45 minutes, the base station does not attempt to restore service-related transmission configurations.
- If a base station is disconnected due to a fault of the bearer network, the base station, when performing the automatic remote OMCH establishment function, cannot restore services immediately after the bearer network recovers. Services can be restored only after the automatic remote OMCH establishment is complete.
- In separate-MPT multimode co-transmission scenarios or base station cascading scenarios, if a base station is disconnected due to a fault of the bearer network, services of other modes or cascaded base stations will be

interrupted when the disconnected base station performs the automatic remote OMCH establishment function.

 NOTE

- These two DHCP detection-based recoveries have the same protocol process. For details, see the DHCP process in *Automatic OMCH Establishment*.
- In IP Layer 3 networking, the DHCP Detect function depends on the DHCP Relay function configured for the transmission bearer network.
- For details about the automatic remote OMCH establishment function, see the chapter of "IP-based Automatic OMCH Establishment for Base Stations" in *Automatic OMCH Establishment*.
- When this function takes effect, the BFD function does not take effect on the base station. If the BFD function of the intermediate transmission device is configured to work in route-binding mode, the route to the base station becomes invalid.
- Automatic remote OMCH establishment is not supported when two UMPT boards work in active/standby mode in IPsec networking.

Automatic Remote OMCH Establishment for eGBTS/NodeB/eNodeB/gNodeB

An eGBTS/NodeB/eNodeB/gNodeB uses either of the preceding two DHCP Detect processes to automatically establish a remote OMCH.

Automatic Remote OMCH Establishment for GBTS

After the OMCH between the GBTS and BSC has been disconnected for 15 minutes, the GBTS initiates a reset. If the OMCH remains disconnected after the reset, the GBTS triggers automatic remote OMCH establishment.

This function is not under switch control and is enabled by default.

5.2 Network Analysis

5.2.1 Benefits

None

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

No requirements

RF Modules

This function does not depend on RF modules.

5.3.4 Others

None

5.4 Operation and Maintenance

This function is enabled on a GBTS by default and is not controlled by a switch. The following sections describe the engineering guidelines for this function on an eGBTS/NodeB/eNodeB/gNodeB.

5.4.1 When to Use

It is recommended that this function be enabled in all scenarios.

The deployment requirements vary depending on the OMCH networking:

- For the deployment requirements in non-IPsec networking, see the sections about configuration requirements for the DHCP server and network devices in *Automatic OMCH Establishment*.
- For the deployment requirements in IPsec networking where the DHCP and OM packets are protected by IPsec and other packets are completely or partially protected by IPsec, see the sections about configuration requirements for the public DHCP server, MAE built-in DHCP server, and network devices in *Automatic OMCH Establishment*.
- For the deployment requirements in IPsec networking where the DHCP packets are not protected by IPsec, OM packets are protected by IPsec, and other packets are completely or partially protected by IPsec, see the sections about configuration requirements for the MAE built-in DHCP server and network devices in *Automatic OMCH Establishment*.
- For the deployment requirements in IPsec networking where the DHCP and OM packets are not protected by IPsec, service and signaling packets are protected by IPsec, and other packets are completely or partially protected by IPsec, see the sections about configuration requirements for the MAE built-in DHCP server and network devices in *Automatic OMCH Establishment*.
- For the deployment requirements in separate-MPT multimode base stations, see the sections about configuration requirements for the DHCP server and network devices in *Automatic OMCH Establishment*.

5.4.2 Precautions

When the startup of an eGBTS/NodeB/eNodeB/gNodeB fails for several consecutive times, the system uses the minimum configuration instead of the current configuration to trigger automatic OM channel establishment.

5.4.3 Data Configuration

5.4.3.1 Data Preparation

MO	Parameter Name	Parameter ID	Setting Notes
DHCPS W	OMCH Auto-Establishment Switch	<i>SWITCH (LTE eNodeB, 5G gNodeB)</i>	Set this parameter to ENABLE(Enable) .

5.4.3.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
SET DHCP SW: SWITCH=ENABLE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
SET DHCP SW: SWITCH=DISABLE;
```

5.4.3.3 Using the MAE-Deployment

- Fast batch activation
This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Feature Operation and Maintenance**.
- Single/Batch configuration
This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4 Activation Verification

Recovery process of all OM channel-related parameters:

Run the MML command **LST DHCP SW** to query the value of **OMCH Auto-Establishment Switch**. If the value of this parameter is **ENABLE**, the function has been enabled.

5.4.5 Network Monitoring

None

6 Other Functions

6.1 Automatic Version Rollback

This function is not under switch control and is enabled by default.

Automatic Version Rollback for eGBTS/NodeB/eNodeB/gNodeB

After the software version for a base station has been upgraded, the base station will reset. After the base station has restarted, its software version will be rolled back to the source version for only once and ALM-26255 Automatic Version Rollback will be reported if one of the following faults occurs:

- The base station detects that the initialization of the current version fails.
- The base station detects that the OM channel of the current version has not been successfully connected, and the interruption has lasted 30 minutes.

After the hot patch of a base station software version takes effect, if one of the following faults occurs, the hot patch will be rolled back to the original state for only once and ALM-26255 Automatic Version Rollback will be reported:

- The base station abnormally resets within ten minutes after the hot patch is activated.
- The OM channel is continuously disconnected within ten minutes after the hot patch is activated, and the base station resets before the OM channel recovery process starts (for example, the transmission link fault rectification function is triggered).

Automatic Version Rollback for GBTS

After the software version for a base station has been upgraded, the base station will reset. After the base station has restarted, its software version will be rolled back to the source version for only once and ALM-26255 Automatic Version Rollback will be reported if one of the following faults occurs:

- The base station detects that the initialization of the current version fails.
- The base station detects that the OM channel of the current version has not been successfully connected, and the interruption has lasted 30 minutes.

The GBTS does not support automatic version rollback of hot patches.

6.2 Transmission Link Fault Rectification

Transmission Link Fault Rectification for eGBTS/NodeB/eNodeB/gNodeB

This function enables a base station to initiate a reset when both the OM channel and services are interrupted. The reset helps resolve certain issues and avoid onsite handling.

If the OM channel and services have been interrupted for 30 or 180 minutes (only in the scenario where the base station starts DHCP detection after a false deletion of the **OMCH MO**), the base station automatically resets to rectify the fault. If the fault persists, the base station will initiate another reset after an interval of 480 minutes. If the two resets still cannot rectify the fault, the base station will not reset again.

NOTE

- The OM channel and service interruption duration (30, 180, or 480 minutes) that triggers an automatic reset will be prolonged by a random value of 0 to 10 minutes. The time when an automatic reset is triggered cannot be manually configured.
- If the local maintenance connection is normal, the base station will not perform transmission link fault rectification. If the local maintenance connection is interrupted later, and both the OM channel and services are interrupted, the base station resets immediately.
- In separate-MPT multimode co-transmission scenarios or base station cascading scenarios, services of other modes or cascaded base stations will be interrupted when an involved base station performs the transmission link fault rectification.

Transmission Link Fault Rectification for GBTS

After the OM channel between the GBTS and BSC has been interrupted for 15 minutes, the GBTS initiates a reset to rectify the fault, which helps avoid onsite handling.

This function is not under switch control and is enabled by default.

NOTE

In this function, the GBTS can reset for an infinite number of times.

6.3 Automatic Recovery of OMCH Running Data

Automatic Recovery of OMCH Running Data for eGBTS/NodeB/eNodeB/gNodeB

This function repairs the OM channel using the running data when the OM channel is steadily connected, instead of using the data obtained from the remote DHCP server. If the automatic recovery fails, the current configurations keep taking effect on the base station. The automatic recovery of OMCH running data function succeeds only in certain scenarios. You are advised to run the **CFM CB** or **MOD CB** command to enable automatic configuration data rollback before

remotely modifying the OMCH-related parameters. For details, see [4 Automatic Configuration Data Rollback](#).

The recovery is classified into the following scenarios:

- Assume that services are running properly and that the OM channel is not protected by IPsec. Without affecting services, the base station can try to restore the Ethernet port (physical-layer parameter), the VLAN (link-layer parameter), as well as the local IP address and route (network-layer parameters) of the OM channel. If the OM channel is protected by IPsec, the base station can try to restore the Ethernet port, the local IP address and route, and the ACL rules of the OM channel.
- When services are not running properly, the base station disables the configurations related to transmission and certificates. In this case, the base station attempts to restore the running data based on the related parameters in the result of automatic OMCH establishment. (The related parameters can be queried by running the **DSP DHCPRSLT** command.) The parameters that can be restored include those on the physical layer (Ethernet port), link layer (VLAN used by the OM channel), and network layer (such as the local IP address and route of the OM channel).

This function is triggered in the following scenarios:

- If the OM channel is interrupted due to configuration modification, this function is triggered to recover the OM channel.
After some IPsec configuration parameters are modified, the modification takes effect when the next negotiation between the base station and SeGW succeeds. At this time, the OMCH may be disconnected. These IPsec parameters include:
 - ENCALG
 - AUTHALG
 - DHGRP
 - PRFALG
 - ENCAPMODE
 - TRANMODE
 - ESPAUTHALG
 - ESPENCALG
- The base station enters the minimum configuration state (that is, the base station runs using the minimum configuration file released with the NE software). This function can be triggered after the base station resets but data cannot be obtained from the DHCP server.

Restrictions in IPsec networking:

- When this function takes effect, IPsec Redundancy Among Multiple SeGWs allows the base station to set up IPsec tunnels to the primary and secondary SeGWs, but IPsec tunnel backup is not supported. For details, see *IPsec*.
- This function supports IPsec networking for transmission ports with **INTERFACE.PT (LTE eNodeB, 5G gNodeB)** set to **ETH**.
- This function supports only the IKE and IPsec negotiation algorithm suites used for establishing a temporary IPsec tunnel during automatic OMCH establishment. For details, see *Automatic OMCH Establishment*.

- In scenarios where certificates are used for IPsec authentication, this function only supports the certificate of the board where IKE is configured. If such a certificate does not exist, the certificate can only be downloaded through the interface IP used by IKE for transmission.
- This function does not support the tunnel-interface-based IPv4 IPsec configuration mode based on dynamic DHCP addresses.

This function supports single-hop BFD scenarios, but with the following restrictions:

- The local IP address of the BFD session must be the same as the OMCH interface IP address for outgoing transmission.
- A maximum of eight sessions are supported.
- BFD authentication is not supported.
- One-arm BFD Echo is not supported.

NOTE

When a user has logged in to the base station, for example, through Web LMT, this function does not take effect.

When this function takes effect and IPsec Redundancy Among Multiple SeGWs is enabled, the IPsec tunnel switchover takes a certain period of waiting time, which may prolong the time required for automatic OMCH recovery. For a single secondary SeGW, the waiting time is related to the sum of **IKECFG.IPSECSBWAITTIME (5G gNodeB, LTE eNodeB)** and **IKECFG.IPSECSORANDTIME (5G gNodeB, LTE eNodeB)**. The more secondary SeGWs are configured for the base station, the longer it takes for automatic OMCH recovery. For details about IPsec Redundancy Among Multiple SeGWs, see *IPsec*.

Risks and impact:

- Only the running data of the OMCH of the base station is restored, and the configuration data of the OMCH is not restored. After data restoration is complete, the running data is inconsistent with the configuration data. In this case, the base station reports ALM-25902 Remote Maintenance Link Running Data and Configuration Mismatch. Services can be restored only after the alarm is cleared. For details about how to clear the alarm, see *3900 & 5900 Series Base Station Alarm Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- When services are running properly, the base station reports ALM-26245 Configuration Data Inconsistency during recovery of the physical-layer parameters of the Ethernet port. For details about how to clear the alarm, see *3900 & 5900 Series Base Station Alarm Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- If the value of **BS Running State** is **OMCH auto-establish is running**, which can be queried by running the **DSP DHCPRLT** command, the base station checks whether service-related transmission configurations need to be restored every 45 minutes.
 - If no transmission-related O&M operations are performed within 45 minutes, the base station may attempt to restore service-related transmission configurations. During this process, the OM channel will be temporarily interrupted.
 - If transmission-related O&M operations are performed within 45 minutes, the base station does not attempt to restore service-related transmission configurations.

- If a base station is disconnected due to incorrect parameter configurations related to multiple OM channels and **BS Running State** is **NORMAL** after automatic recovery of OMCH running data, it is recommended that the parameter configurations be corrected in batches (for example, by activating the configuration file). Otherwise, running a single MML command to correct the parameter configurations one by one, the base station may be disconnected again during the operation.
- If the base station is disconnected due to a fault on the bearer network, the base station cannot restore services immediately after the fault is rectified. Services can be restored only after the automatic recovery of the OMCH running data is complete.
- In separate-MPT multimode co-transmission scenarios or base station cascading scenarios, if a base station is disconnected due to a bearer network fault, services of other modes or cascaded base stations will be interrupted when the disconnected base station uses the function of automatic recovery of OMCH running data.
- During the automatic recovery, the OM IP address and interface IP address of the OMCH in the normal state are used. If either of these two IP addresses has been allocated to another device, this function will incur an IP address conflict.

Automatic Recovery of OMCH Running Data for GBTS

The GBTS does not support this function.

6.4 OM Emergency Channel-based Self-Healing

eGBTS/NodeB/eNodeB/gNodeB OM Emergency Channel-based Self-Healing

When the OMCH of the base station remains interrupted for a period of time, the base station reports the SCTP link status to the MAE through the OM emergency channel. Users can determine whether to enable OM emergency channel-based self-healing based on the SCTP link status. If users select **Enable**, the MAE enables OM emergency channel-based self-healing for 2 hours and then disables the channel automatically. During this period, if the base station sends an OM channel self-healing request again, the MAE responds with an acknowledgment message. The base station forcibly resets after receiving the acknowledgment message from the MAE. If the OM channel recovers after the base station is reset, the OM emergency channel-based self-healing is complete. Otherwise, the base station enters the minimum configuration mode and starts the automatic OM channel establishment procedure. For details, see *Automatic OMCH Establishment*.

To enable the OM emergency channel-based self-healing function, perform the following operations on the base station and MAE sides:

- Step 1** On the base station side: Select the **OMCH_ESCAPE_SW** option of the **TRANSALGEXTPARA.OMCHMNGSW (LTE eNodeB, 5G gNodeB)** parameter. This option is selected by default. To disable this function, run the **SET TRANSALGEXTPARA** command with the **OMCH_ESCAPE_SW** option of the **TRANSALGEXTPARA.OMCHMNGSW (LTE eNodeB, 5G gNodeB)** parameter deselected. After the OM emergency channel is disabled on the base station, OM emergency channel-based self-healing cannot be enabled through the MAE.

Step 2 On the MAE side: Log in to the MAE-Access, choose **FMA > Transmission > Fault Recovery Confirmation > OM Emergency Channel (OM Link Disconnection)**, and click **Query**. Select the base station for which OM emergency channel-based self-healing needs to be enabled and click **Enable**.

----End

The networking restrictions are as follows:

This function applies only to the scenario where the OM channel is transmitted through the active main control board, the **INTERFACE.PT (LTE eNodeB, 5G gNodeB)** parameter is set to **ETH**, and the **INTERFACE.ITFTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **VLAN**.

This function does not apply to networks where OMCH data is protected by IPsec.

This function is not supported when the base station software running mode is set to **EMERGENCYMODE**.

 CAUTION

- If the OM channel of the base station is interrupted for more than 40 minutes, the base station reports the SCTP link status to the MAE through the OM emergency channel. If the SCTP link status cannot be queried on the MAE, you are advised to perform troubleshooting on the intermediate transport network or contact Huawei technical support.
 - Enabling the OM emergency channel (OM link disconnection) is a high-risk operation, which will cause the base station to reset or enter the minimum configuration mode. Perform this operation under the guidance of Huawei engineers.
-

GBTS OM Emergency Channel-based Self-Healing

The GBTS does not support this function.

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- *SRAN Networking and Evolution Overview*
- *Automatic OMCH Establishment*
- *3900 & 5900 Series Base Station Initial Configuration Guide in 3900 & 5900 Series Base Station Product Documentation*